GEOPHYSICAL SURVEY REPORT

sumo Survey

GEOPHYSICS FOR ARCHAEOLOGY & ENGINEERING

Land at South Marston, Swindon Eastern Village

Client
Orion Heritage

Survey Report
SOR10797

Date
February 2017

Incorporating

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and

STRATASCAN LTD

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Project name:
Land at South Marston, Swindon Eastern Village

SUMO Job reference:
SOR10797

Client:
Orion Heritage

Survey date:
16-18 January 2017

Report date:
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DIGITAL CONTENT (Archive Data)

- Minimally Processed Greyscale Images, Colour Plots and XY Trace Plots in DWG format
- DWG Viewer
- Digital Copies of Report Text and Figures (both PDF and native formats)

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1 SUMMARY OF RESULTS

A detailed magnetic survey was carried out over approximately 11.2 hectares of arable land and grassland. No archaeological anomalies have been detected. A discrete response and linear anomalies of uncertain origin have been identified, along with areas of natural variation and modern ferrous responses.

2 INTRODUCTION

2.1 Background synopsis

SUMO were commissioned to undertake a geophysical survey of an area outlined for development. This survey forms part of an archaeological investigation being undertaken by **Orion Heritage.**

2.2 Site details

NGR / Postcode	SU 215 874/ SN6 8HS
Location	The site is located to the south-west of Shrivenham, Oxfordshire. The site is bound to the south by the A420, and to the east by the River Cole.
Unitary Authority	Swindon
Parish	Highworth CP
Topography	Mostly flat
Current Land Use	Arable/grassland
Weather Conditions	Cold/ wet
Geology	Solid: Sandstone of Down Sand Member is recorded across the north with mudstone of Ampthill Clay Formation and Kimmeridge Clay Formation (undifferentiated) to the south of the site. Superficial: Alluvial deposits of clay, silt, sand and gravel are recorded along the eastern edge, following the course of the River Cole (British Geological Survey website).
Soils	The soils across the majority of the site are of the Burseledon Association (572j), with soils of the Fladbury 1 Association (813b) along the eastern edge (Soil Survey of England and Wales, Sheet 6, South East England).
Archaeology	A search of Oxfordshire HER (2017) within a 1km radius of the site identifies a possible prehistoric pit alignment (HER No. 12059), on the eastern side of the River Cole. No other archaeological remains are recorded in the area.
Survey Methods	Magnetometer survey (fluxgate gradiometer)
Study Area	11.2 ha

2.3 Aims and Objectives

To locate and characterise any anomalies of possible archaeological interest within the study area.

3 METHODS, PROCESSING & PRESENTATION

3.1 Standards & Guidance

This report and all fieldwork have been conducted in accordance with the latest guidance documents issued by Historic England (EH 2008) (then English Heritage) and the Chartered Institute for Archaeologists (IfA 2002 & CIfA 2014).

3.2 Survey methods

Detailed magnetic survey was chosen as an efficient and effective method of locating archaeological anomalies.

Technique	Instrument	Traverse Interval	Sample Interval
Magnetometer	Bartington Grad 601-2	1.0m	0.25m

More information regarding this technique is included in Appendix A

3.3 Data Processing

The following basic processing steps have been carried out on the data used in this report:

De-stripe
De-stagger
Interpolate

3.4 Presentation of results and interpretation

. The presentation of the results for each site involves a grey-scale and colour-scale plot of processed data. Magnetic anomalies are identified, interpreted and plotted onto the 'Interpretation' drawings. The minimally processed data are provided as a greyscale image in the Archive Data Folder with an XY trace plot in CAD format. A CAD viewer is also provided.

When interpreting the results, several factors are taken into consideration, including the nature of archaeological features being investigated and the local conditions at the site (geology, pedology, topography etc.). Anomalies are categorised by their potential origin. Where responses can be related to other existing evidence, the anomalies will be given specific categories, such as: *Abbey Wall* or *Roman Road*. Where the interpretation is based largely on the geophysical data, levels of confidence are implied, for example: *Probable*, or *Possible Archaeology*. The former is used for a confident interpretation, based on anomaly definition and/or other corroborative data such as cropmarks. Poor anomaly definition, a lack of clear patterns to the responses and an absence of other supporting data reduces confidence, hence the classification *Possible*.

4 RESULTS

*The survey covers a single area. Where considered appropriate, specific anomalies have been given numerical labels **[1] [2]** and these appear in the text below, as well as on the Interpretation Figure(s).*

4.1 *Probable/Possible Archaeology*

No magnetic responses have been recorded that could be interpreted as being of probable or possible archaeological interest.

4.4 *Agricultural – Ploughing, Land drains*

No magnetic responses characteristic of agricultural activity have been identified.

4.5 *Natural / Geological / Pedological / Topographic*

Sinuous bands of enhanced magnetic variation in the north of the site are characteristic of responses detected near to watercourses, and may relate to underlying alluvial deposits or paleo-channels.

4.6 *Uncertain*

A small, discrete positive anomaly **[1]** in the south of the site may be associated with a shallow, backfilled pit, though may equally be of natural origin.

Very weak, parallel linear anomalies **[2]** in the north-east are indicative of former cut features, such as ditches. The exact origin of these responses is unclear, and they may be archaeological or of later agricultural origin.

Further weak linear responses are of uncertain origin and may be agricultural or natural.

4.7 *Ferrous / Magnetic Disturbance*

A small area of magnetic debris in the north-west of the site is indicative of modern ferrous rubbish in the topsoil.

Ferrous responses close to boundaries are due to adjacent fences and gates. Smaller scale ferrous anomalies ("iron spikes") are present throughout the data and their form is best illustrated in the XY trace plots or the colour-scale plots. These responses are characteristic of small pieces of ferrous debris (or brick / tile / igneous rocks) in the topsoil and are commonly assigned a modern origin. Only the most prominent of these are highlighted on the interpretation diagram.

5 DATA APPRAISAL & CONFIDENCE ASSESSMENT

English Heritage Guidelines (EH 2008) Table 4 states that the average magnetic response over mudstone geologies is variable. However, the detection of linear anomalies of uncertain origin, suggests that this survey is likely to have detected any archaeological features, if present.

6 CONCLUSION

The survey at South Marston has not identified any archaeological features. The anomalies detected include a small discrete response and linears of uncertain origin, areas of natural magnetic variation and magnetic disturbance from nearby ferrous objects.

7 REFERENCES

BGS 2017 British Geological Survey *website*: (http://www.bgs.ac.uk/opengeoscience/home.html?Accordion1=1#maps) Geology of Britain viewer [Accessed 06/02/2017].

CIfA 2014 *Standard and Guidance for Archaeological Geophysical Survey.* Amended 2016. CIfA Guidance note. Chartered Institute for Archaeologists, Reading http://www.archaeologists.net/sites/default/files/CIfAS%26GGeophysics_2.pdf

EH 2008 *Geophysical Survey in Archaeological Field Evaluation.* English Heritage, Swindon https://content.historicengland.org.uk/images-books/publications/geophysical-survey-in-archaeological-field-evaluation/geophysics-guidelines.pdf/

OCC Oxfordshire County Council. *Historic Environment Record [online]* Available through: www.heritagegateway.org.uk [Accessed 07/02/2017].

IfA 2002 *The Use of Geophysical Techniques in Archaeological Evaluations*, IFA Paper No 6, C. Gaffney, J. Gater and S. Ovenden. Institute for Archaeology, Reading

SSEW 1983 *Soils of England and Wales. Sheet 6, South East England.* Soil Survey of England and Wales, Harpenden.

Appendix A - Technical Information: Magnetometer Survey Method

Grid Positioning
For hand held gradiometers the location of the survey grids has been plotted together with the referencing information. Grids were set out using a Trimble R8 Real Time Kinematic (RTK) VRS Now GNSS GPS system.

An RTK GPS (Real-time Kinematic Global Positioning System) can locate a point on the ground to a far greater accuracy than a standard GPS unit. A standard GPS suffers from errors created by satellite orbit errors, clock errors and atmospheric interference, resulting in an accuracy of 5m-10m. An RTK system uses a single base station receiver and a number of mobile units. The base station re-broadcasts the phase of the carrier it measured, and the mobile units compare their own phase measurements with those they received from the base station. This results in an accuracy of around 0.01m.

Technique	Instrument	Traverse Interval	Sample Interval
Magnetometer	Bartington Grad 601-2	1m	0.25m

Instrumentation: Bartington *Grad* 601-2
Bartington instruments operate in a gradiometer configuration which comprises fluxgate sensors mounted vertically, set 1.0m apart. The fluxgate gradiometer suppresses any diurnal or regional effects. The instruments are carried, or cart mounted, with the bottom sensor approximately 0.1-0.3m from the ground surface. At each survey station, the difference in the magnetic field between the two fluxgates is measured in nanoTesla (nT). The sensitivity of the instrument can be adjusted; for most archaeological surveys the most sensitive range (0.1nT) is used. Generally, features up to 1m deep may be detected by this method, though strongly magnetic objects may be visible at greater depths. The Bartington instrument can collect two lines of data per traverse with gradiometer units mounted laterally with a separation of 1.0m. The readings are logged consecutively into the data logger which in turn is daily down-loaded into a portable computer whilst on site. At the end of each site survey, data is transferred to the office for processing and presentation.

Data Processing

Zero Mean Traverse	This process sets the background mean of each traverse within each grid to zero. The operation removes striping effects and edge discontinuities over the whole of the data set.
Step Correction (De-stagger)	When gradiometer data are collected in 'zig-zag' fashion, stepping errors can sometimes arise. These occur because of a slight difference in the speed of walking on the forward and reverse traverses. The result is a staggered effect in the data, which is particularly noticeable on linear anomalies. This process corrects these errors.

Display

Greyscale/ Colourscale Plot	This format divides a given range of readings into a set number of classes. Each class is represented by a specific shade of grey, the intensity increasing with value. All values above the given range are allocated the same shade (maximum intensity); similarly, all values below the given range are represented by the minimum intensity shade. Similar plots can be produced in colour, either using a wide range of colours or by selecting two or three colours to represent positive and negative values. The assigned range (plotting levels) can be adjusted to emphasise different anomalies in the data-set.

Interpretation Categories

In certain circumstances (usually when there is corroborative evidence from desk-based or excavation data) very specific interpretations can be assigned to magnetic anomalies (for example, *Roman Road, Wall,* etc.) and where appropriate, such interpretations will be applied. The list below outlines the generic categories commonly used in the interpretation of the results.

Archaeology / Probable Archaeology	This term is used when the form, nature and pattern of the responses are clearly or very probably archaeological and /or if corroborative evidence is available. These anomalies, whilst considered anthropogenic, could be of any age.
Possible Archaeology	These anomalies exhibit either weak signal strength and / or poor definition, or form incomplete archaeological patterns, thereby reducing the level of confidence in the interpretation. Although the archaeological interpretation is favoured, they may be the result of variable soil depth, plough damage or even aliasing as a result of data collection orientation.
Industrial / Burnt-Fired	Strong magnetic anomalies that, due to their shape and form or the context in which they are found, suggest the presence of kilns, ovens, corn dryers, metal-working areas or hearths. It should be noted that in many instances modern ferrous material can produce similar magnetic anomalies.
Former Field Boundary (probable & possible)	Anomalies that correspond to former boundaries indicated on historic mapping, or which are clearly a continuation of existing land divisions. Possible denotes less confidence where the anomaly may not be shown on historic mapping but nevertheless the anomaly displays all the characteristics of a field boundary.
Ridge & Furrow	Parallel linear anomalies whose broad spacing suggests ridge and furrow cultivation. In some cases, the response may be the result of more recent agricultural activity.
Agriculture (ploughing)	Parallel linear anomalies or trends with a narrower spacing, sometimes aligned with existing boundaries, indicating more recent cultivation regimes.
Land Drain	Weakly magnetic linear anomalies, quite often appearing in series forming parallel and herringbone patterns. Smaller drains may lead and empty into larger diameter pipes, which in turn usually lead to local streams and ponds. These are indicative of clay fired land drains.
Natural	These responses form clear patterns in geographical zones where natural variations are known to produce significant magnetic distortions.
Magnetic Disturbance	Broad zones of strong dipolar anomalies, commonly found in places where modern ferrous or fired materials (e.g. brick rubble) are present. They are presumed to be modern.
Service	Magnetically strong anomalies, usually forming linear features are indicative of ferrous pipes/cables. Sometimes other materials (e.g. pvc) or the fill of the trench can cause weaker magnetic responses which can be identified from their uniform linearity.
Ferrous	This type of response is associated with ferrous material and may result from small items in the topsoil, larger buried objects such as pipes, or above ground features such as fence lines or pylons. Ferrous responses are usually regarded as modern. Individual burnt stones, fired bricks or igneous rocks can produce responses similar to ferrous material.
Uncertain Origin	Anomalies which stand out from the background magnetic variation, yet whose form and lack of patterning gives little clue as to their origin. Often the characteristics and distribution of the responses straddle the categories of *Possible Archaeology* / *Natural* or (in the case of linear responses) *Possible Archaeology* / *Agriculture*; occasionally they are simply of an unusual form.

Where appropriate some anomalies will be further classified according to their form (positive or negative) and relative strength and coherence (trend: weak and poorly defined).

Appendix B - Technical Information: Magnetic Theory

Detailed magnetic survey can be used to effectively define areas of past human activity by mapping spatial variation and contrast in the magnetic properties of soil, subsoil and bedrock. Although the changes in the magnetic field resulting from differing features in the soil are usually weak, changes as small as 0.1 nanoTeslas (nT) in an overall field strength of 48,000 (nT), can be accurately detected.

Weakly magnetic iron minerals are always present within the soil and areas of enhancement relate to increases in *magnetic susceptibility* and permanently magnetised *thermoremanent* material.

Magnetic susceptibility relates to the induced magnetism of a material when in the presence of a magnetic field. This magnetism can be considered as effectively permanent as it exists within the Earth's magnetic field. Magnetic susceptibility can become enhanced due to burning and complex biological or fermentation processes.

Thermoremanence is a permanent magnetism acquired by iron minerals that, after heating to a specific temperature known as the Curie Point, are effectively demagnetised followed by re-magnetisation by the Earth's magnetic field on cooling. Thermoremanent archaeological features can include hearths and kilns; material such as brick and tile may be magnetised through the same process.

Silting and deliberate infilling of ditches and pits with magnetically enhanced soil creates a relative contrast against the much lower levels of magnetism within the subsoil into which the feature is cut. Systematic mapping of magnetic anomalies will produce linear and discrete areas of enhancement allowing assessment and characterisation of subsurface features. Material such as subsoil and non-magnetic bedrock used to create former earthworks and walls may be mapped as areas of lower enhancement compared to surrounding soils.

Magnetic survey is carried out using a fluxgate gradiometer which is a passive instrument consisting of two sensors mounted vertically 1m apart. The instrument is carried about 30cm above the ground surface and the top sensor measures the Earth's magnetic field whilst the lower sensor measures the same field but is also more affected by any localised buried feature. The difference between the two sensors will relate to the strength of a magnetic field created by this feature, if no field is present the difference will be close to zero as the magnetic field measured by both sensors will be the same.

Factors affecting the magnetic survey may include soil type, local geology, previous human activity and disturbance from modern services.

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Licence No: AL 50125A
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Stratascan Ltd.
Vineyard House
Upper Hook Road
Upton Upon Severn
WR8 0SA

OS 100km square = SU

N

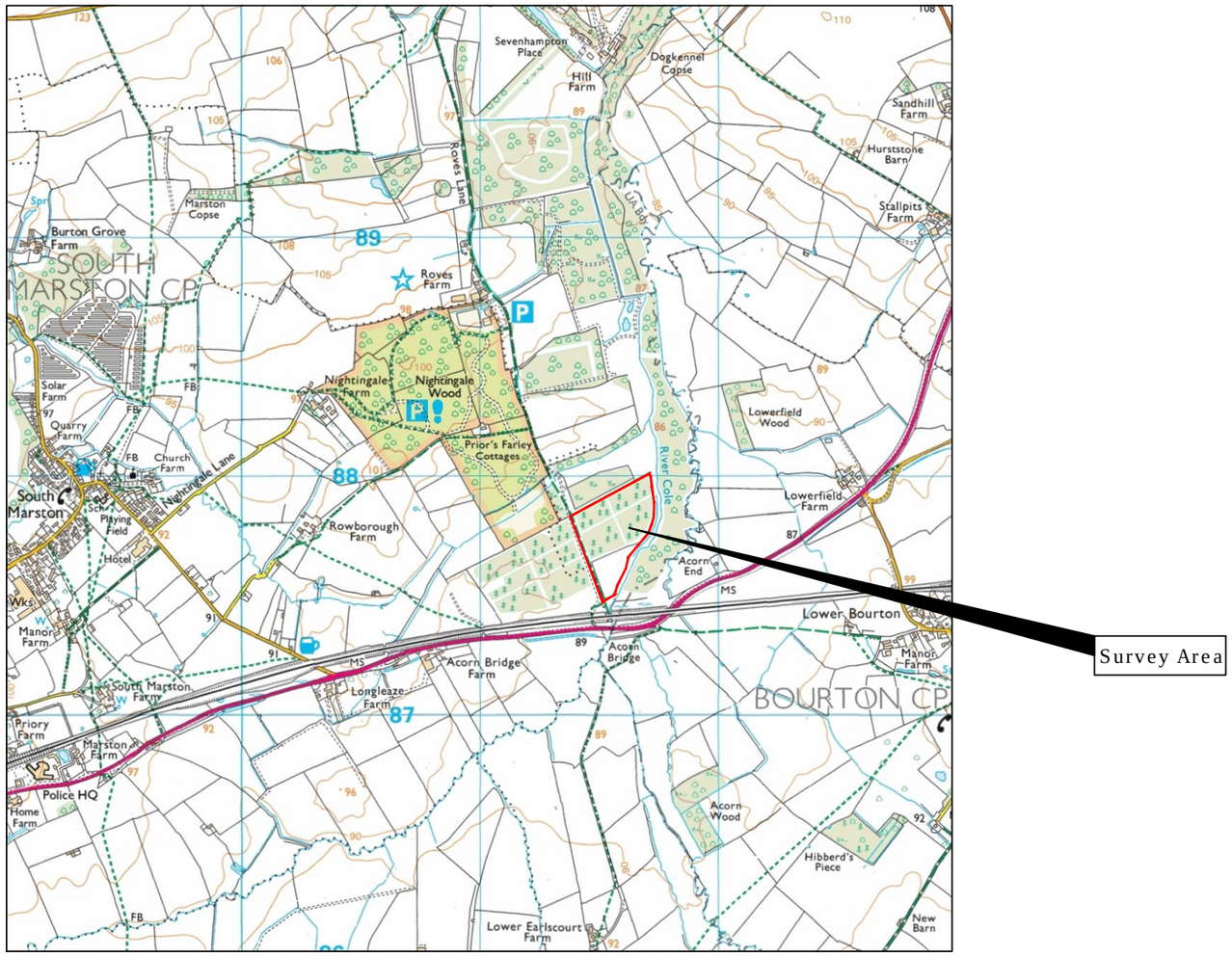
90

89

88

87

86



19 20 21 22 23

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GEOPHYSICS FOR ARCHAEOLOGY & ENGINEERING

Title:
LOCATION PLAN OF SURVEY AREA

Client:
ORION HERITAGE

Project:
SOR10797 - LAND AT SOUTH MARSTON, SWINDON EASTERN VILLAGE

Scale: 1:25000 @ A3
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Fig No:
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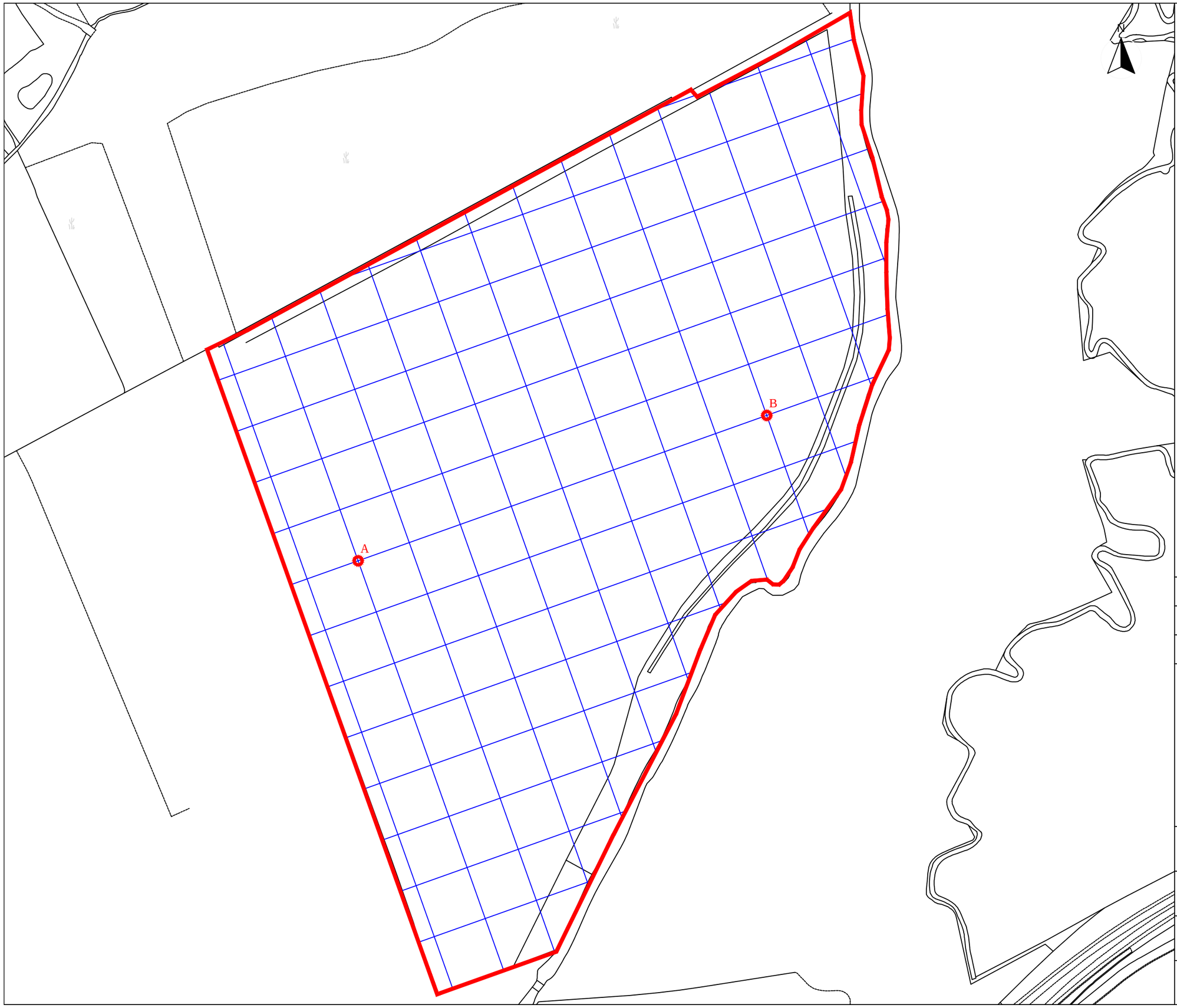

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GEOPHYSICS FOR ARCHAEOLOGY & ENGINEERING

Title:
REFERENCING

Client:
ORION HERITAGE

Project:
SOR10797 - LAND AT SOUTH MARSTON, SWINDON EASTERN VILLAGE

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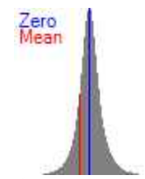
Minimum -1nT (white)



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+2nT

-1nT

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Title:
PLOT OF MINIMALLY PROCESSED GRADIOMETER DATA

Client:
ORION HERITAGE

Project:
SOR10797 - LAND AT SOUTH MARSTON, SWINDON EASTERN VILLAGE

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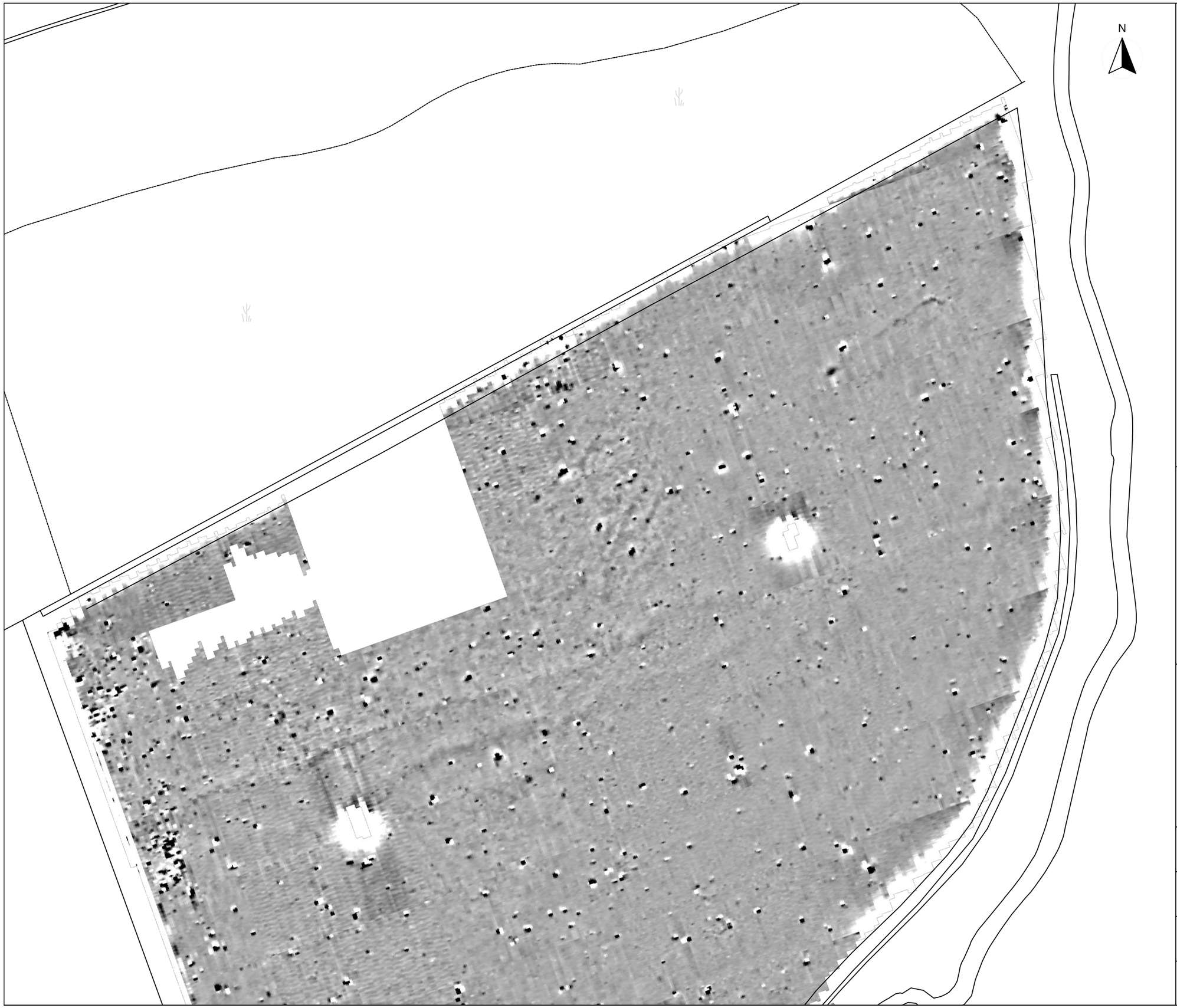

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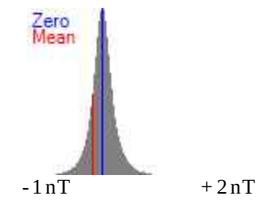

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Title:
PLOT OF MINIMALLY PROCESSED GRADIOMETER DATA - NORTH

Client:
ORION HERITAGE

Project:
SOR10797 - LAND AT SOUTH MARSTON, SWINDON EASTERN VILLAGE

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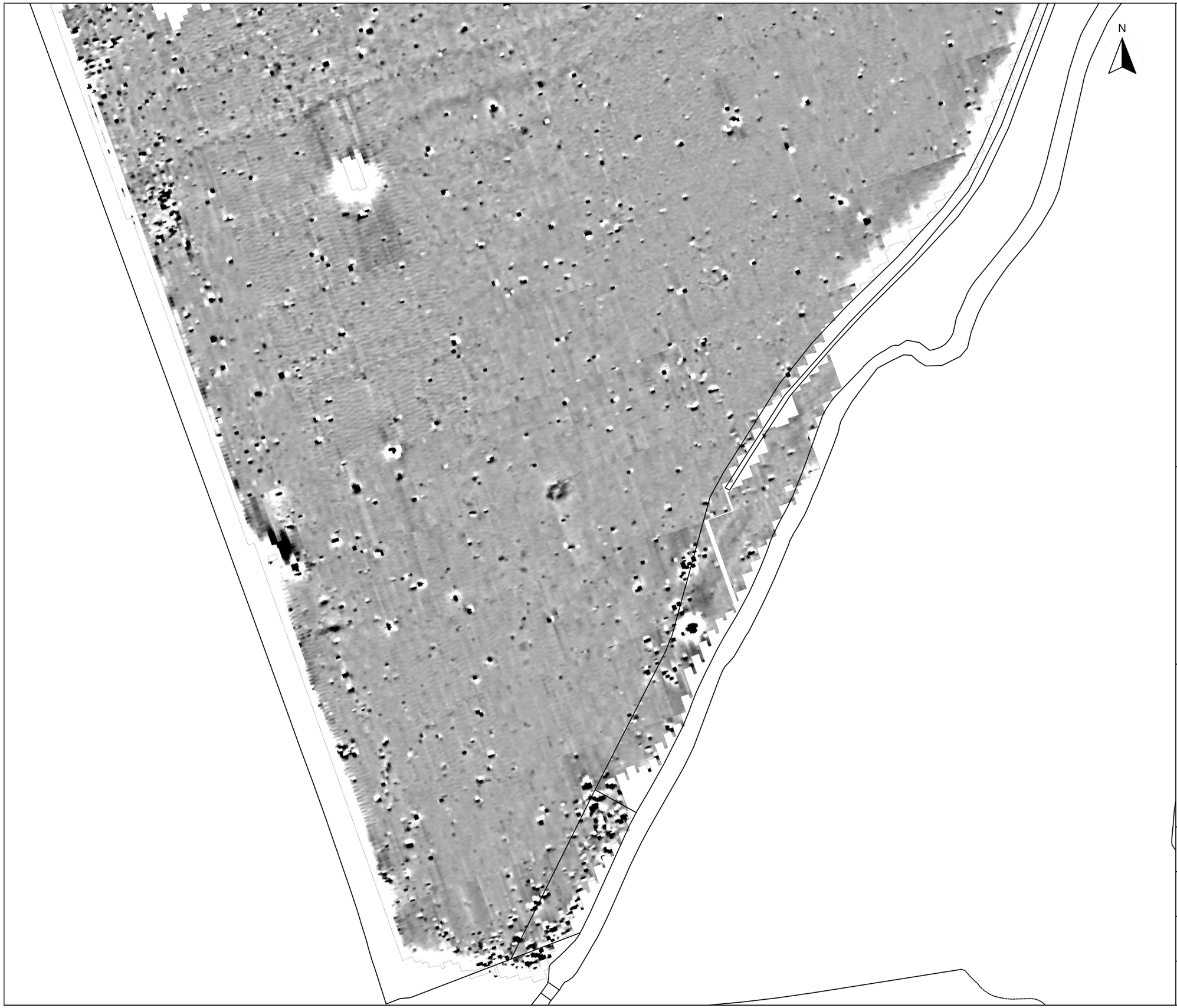

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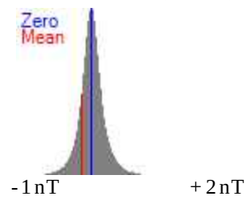

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Title:
PLOT OF MINIMALLY PROCESSED GRADIOMETER DATA - SOUTH

Client:
ORION HERITAGE

Project:
SOR10797 - LAND AT SOUTH MARSTON, SWINDON EASTERN VILLAGE

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Fig No:
05

KEY

	Natural (e.g. geological or pedological)
	Magnetic disturbance
	Ferrous
	Uncertain positive/area of increased response/negative/trend

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GEOPHYSICS FOR ARCHAEOLOGY & ENGINEERING

Title:
ABSTRACTION AND INTERPRETATION OF GRADIOMETER ANOMALIES

Client:
ORION HERITAGE

Project:
SOR10797 - LAND AT SOUTH MARSTON, SWINDON EASTERN VILLAGE

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Fig No:
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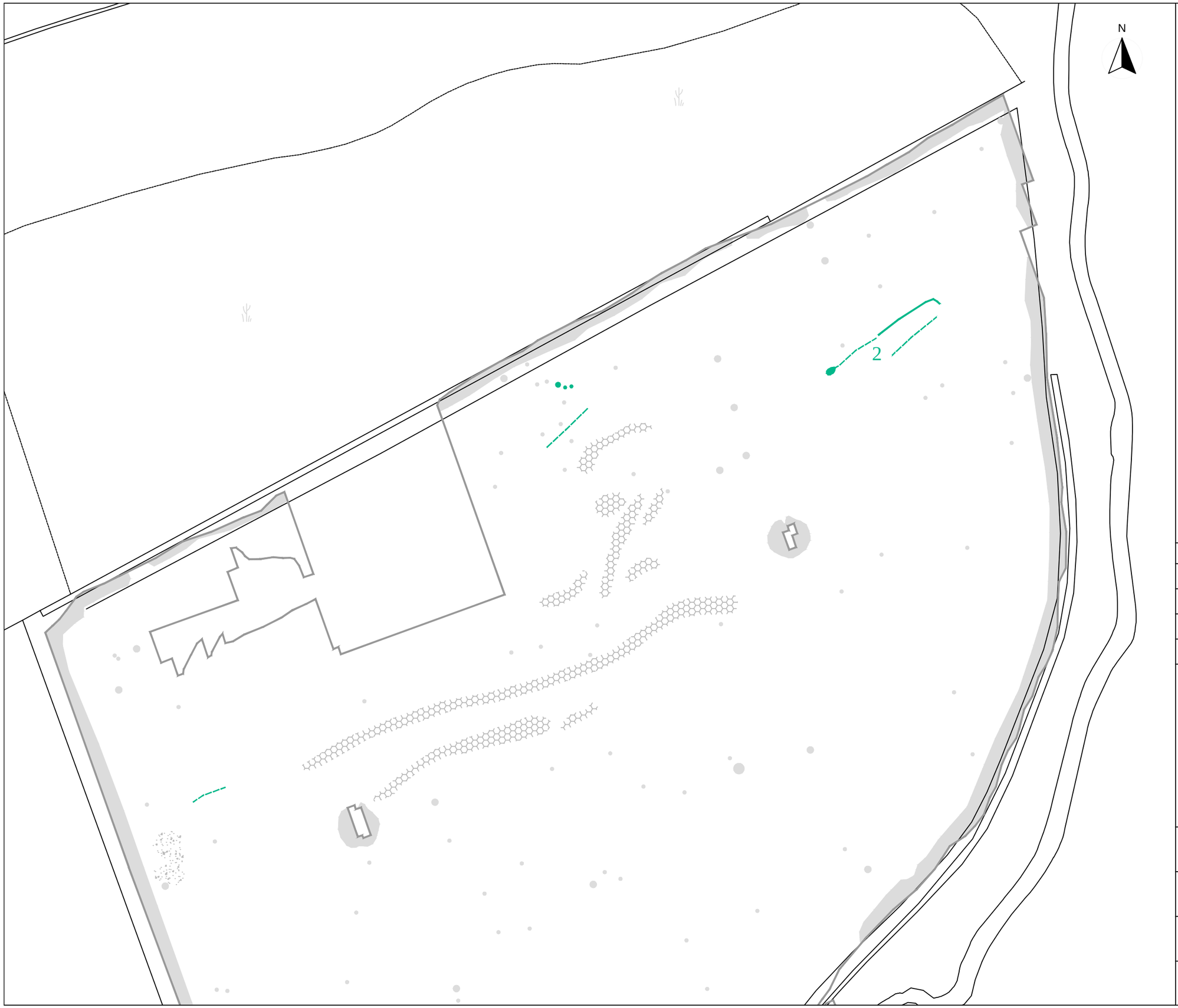

KEY

	Natural (e.g. geological or pedological)
	Magnetic disturbance
	Ferrous
	Uncertain positive/area of increased response/negative/trend

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ABSTRACTION AND INTERPRETATION OF GRADIOMETER ANOMALIES - NORTH

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Scale: 1:1000 @ A3

0m 10 20 30 40 50

Fig No:
07

KEY

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	Magnetic disturbance
	Ferrous
	Uncertain positive/area of increased response/negative/trend

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Title:
ABSTRACTION AND INTERPRETATION OF GRADIOMETER ANOMALIES - SOUTH

Client:
ORION HERITAGE

Project:
SOR10797 - LAND AT SOUTH MARSTON, SWINDON EASTERN VILLAGE

Scale: 1:1000 @ A3

0m 10 20 30 40 50

Fig No:
08

- Archaeological
- Geophysical
- Laser Scanning
- Measured Building
- Topographic
- Utility Mapping

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